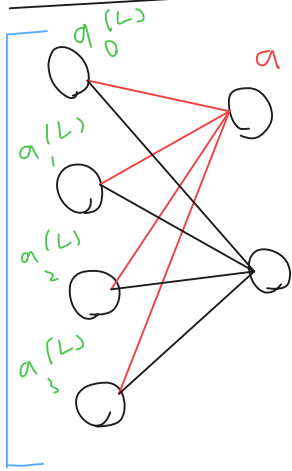


Forward Pass :



$$a_0^{(L+1)} = 0$$

$$\left(\sum_{i=0}^{n_{in}-1} a_i^{(L)} \cdot w_{0,i}^{(L+1)} \right) + b_0^{(L+1)}$$

$(z_0^{(L+1)})$

$\frac{1}{1 + e^{-x}}$ (Sigmoid clamps values between $0 < x < 1$)

$$a = \sigma(z)$$

Input Layer (L)

- → "0"
- → "1"
- → "2"
- ...
-
-
-
-
-
-

Define Cost Function using MSE :

$$C = \sum_{j=0}^{n_{out}-1} (a_j - y_j)^2$$

Expected value of output neuron (e.g., 0 or 1.0)

Smaller cost when predictions are closer to expected values

Output Layer ($\hat{y}$)